Table of Contents

Comprehensive Question Bank: Chapter 9 — Cybersecurity and Safe Digital Collaboration

Section A: Multiple-Choice Questions (MCQs)

Topic 9.1.1: Software Maintenance and Security Patching Mechanics

1. What is a "patch" in software maintenance?
   A) A new operating system
   B) A small update released to fix a specific problem, usually a security flaw
   C) A type of malware
   D) A hardware component

2. Who typically discovers a software vulnerability before a patch is created?
   A) The end user
   B) A security researcher
   C) The internet service provider
   D) The government

3. What term describes a flaw that is already publicly known but has not yet been patched by a specific user?
   A) Zero-day vulnerability
   B) N-day vulnerability
   C) Firewall breach
   D) Ransomware event

4. What describes a flaw that is exploited before any patch exists for it?
   A) N-day vulnerability
   B) Zero-day exploit
   C) Legacy bug
   D) Sandboxed threat

5. What happened as a direct result of the 1988 Morris Worm incident?
   A) The creation of the first antivirus company
   B) The creation of the world's first Computer Emergency Response Team (CERT)
   C) The invention of the firewall
   D) The passing of the GDPR law

6. What is the main danger of leaving software unpatched?
   A) It runs faster
   B) It leaves a known security hole open for attackers to exploit
   C) It automatically deletes files
   D) It improves compatibility

7. Ignoring regular software updates primarily increases the risk of:
   A) Faster performance
   B) Malware infection and data breaches
   C) Extra storage space
   D) Better battery life

8. Robert Morris released his worm in 1988 with what original intention?
   A) To steal financial data
   B) To measure the size of the early internet as an experiment
   C) To destroy government systems
   D) To sell ransomware

Topic 9.1.2: Privacy and Security Settings

1. What is the primary purpose of privacy settings on a digital platform?
   A) To increase advertisement revenue
   B) To let users control who can see their personal information and activity
   C) To make an app run faster
   D) To automatically back up files

2. Why do most apps ship with privacy-weak default settings?
   A) Because it is required by law
   B) Because defaults usually favor convenience and data collection over privacy
   C) Because users demand it
   D) Because it saves battery

3. What does a privacy policy explain to a user?
   A) The app's marketing strategy
   B) How the platform collects, stores, uses, and shares user data
   C) The app's source code
   D) The company's hiring process

4. On Instagram, which setting ensures only approved followers can view your posts?
   A) Two-Factor Authentication
   B) Private account setting
   C) Location sharing
   D) Ad personalization

5. What is the safest default choice for location sharing on a mobile app?
   A) Always on
   B) Off, or "While using app"
   C) Shared with all contacts
   D) Public to everyone

6. On WhatsApp, disabling "read receipts" (blue ticks) primarily affects:
   A) Message encryption
   B) Whether others can see if you have read their messages
   C) Your account password
   D) Your internet speed

7. Which of the following is considered a safer app permission practice?
   A) Granting every app access to contacts, camera, and microphone
   B) Granting only the permissions an app truly needs to function
   C) Turning off all security settings
   D) Sharing your password with the app developer

Topic 9.1.3: Safe Online Collaboration Workflows

1. What does "access control" refer to in online collaboration?
   A) The speed of the internet connection
   B) Deciding who is allowed to open a specific file or folder
   C) The color scheme of an app
   D) The number of devices connected

2. In Google Docs, which permission level should be used if collaborators only need to read a document?
   A) Edit
   B) View only
   C) Owner
   D) Admin

3. What is a "token" in the context of token authentication?
   A) A physical security key
   B) A temporary digital credential proving a user is logged in, used instead of re-entering a password
   C) A type of malware
   D) A firewall rule

4. What is the main risk of an authentication token being stolen?
   A) The device will slow down
   B) An attacker can act as the user without ever knowing their password
   C) The app will crash
   D) The internet bill will increase

5. Which platform is commonly used for secure, protected online collaboration in schools?
   A) Paint
   B) Google Workspace
   C) Notepad
   D) VLC Player

6. What does digital etiquette mean in a shared group chat?
   A) Sending frequent unrelated memes
   B) Communicating politely, respectfully, and staying on-topic
   C) Ignoring other members' opinions
   D) Sharing private information freely

7. Which sharing setting poses the greatest security risk for a collaborative folder?
   A) Restricted access with named collaborators
   B) "Anyone with the link can edit"
   C) View-only access for guests
   D) Password-protected access

Topic 9.2.1: Understanding Cyber Threats

1. What historical event is widely considered a founding moment of computer science related to cryptography?
   A) The invention of the firewall
   B) The breaking of the Enigma code at Bletchley Park
   C) The launch of GDPR
   D) The Morris Worm incident

2. Phishing attacks primarily work by:
   A) Directly hacking into a server's code
   B) Tricking users into voluntarily handing over information through fake messages or websites
   C) Physically stealing a device
   D) Overloading a website with traffic

3. Which type of malware locks a victim's files and demands payment for their release?
   A) Spyware
   B) Ransomware
   C) Adware
   D) Firmware
4. What does spyware do?
   A) Locks files for payment
   B) Quietly watches and records user activity without consent
   C) Floods a server with traffic
   D) Filters network traffic
5. A Man-in-the-Middle (MITM) attack occurs when:
   A) A user forgets their password
   B) An attacker secretly intercepts communication between a user and a website
   C) A firewall blocks harmful traffic
   D) A user installs an update
6. Public, unsecured Wi-Fi networks are particularly risky because they make which type of attack easier?
   A) Ransomware
   B) Man-in-the-Middle (MITM) attacks
   C) E-waste generation
   D) Phishing via mailed letters
7. A DDoS (Distributed Denial of Service) attack works by:
   A) Encrypting a victim's files
   B) Flooding a server with traffic from many computers at once until it collapses
   C) Stealing a single password
   D) Reading private emails
8. Which type of vulnerability is exploited before any patch or public knowledge of it exists?
   A) N-day vulnerability
   B) Zero-day exploit
   C) Legacy vulnerability
   D) Sandbox vulnerability
9. Stuxnet (2010) was significant because it proved that:
   A) Firewalls are unnecessary
   B) Malicious software can cause physical damage to real-world industrial infrastructure
   C) Ransomware only affects personal computers
   D) Antivirus software is obsolete

10. Which cyberattack is best described as "a fake delivery driver in uniform asking for your house keys"?
    A) DDoS attack
    B) Phishing
    C) Sandboxing
    D) Hashing

Topic 9.2.2: Security Tools and Techniques

1. What is the primary function of a firewall?
   A) To scan files for known viruses
   B) To monitor and control incoming and outgoing network traffic based on security rules
   C) To encrypt stored passwords
   D) To back up user data

2. What does an Intrusion Detection System (IDS) do?
   A) Actively blocks all network traffic
   B) Monitors a network for suspicious activity and alerts users
   C) Deletes infected files automatically
   D) Encrypts outgoing messages

3. Which analogy best describes the combined role of a firewall and an IDS?
   A) A locked door with no guard
   B) A security guard checking IDs at the gate, plus cameras watching for suspicious behavior inside
   C) A single antivirus scan
   D) A backup power generator

4. What is the main purpose of antivirus software?
   A) To create encrypted tunnels
   B) To detect, block, and remove known viruses and harmful software from a device
   C) To manage user permissions
   D) To generate strong passwords

5. Multi-Factor Authentication (MFA) improves security primarily by:
   A) Making passwords shorter
   B) Requiring two or more independent proofs of identity before granting access
   C) Removing the need for passwords entirely
   D) Increasing internet speed

6. What is "sandboxing" in cybersecurity?
   A) Permanently deleting suspicious files
   B) Running an unknown program in an isolated environment to contain potential harm
   C) Encrypting all network traffic
   D) Blocking all incoming emails
7. Biometric verification uses which of the following to confirm identity?
   A) A memorized password
   B) Physical features such as fingerprints, face scans, or voice patterns
   C) A mailed security code
   D) A firewall rule
8. Which security layer is best described as "checking IDs at a building gate"?
   A) Antivirus
   B) Firewall
   C) Hashing
   D) Sandboxing

Topic 9.2.3: Data Protection and Cryptography

1. What does encryption do to readable data?
   A) Deletes it permanently
   B) Converts it into unreadable coded text (ciphertext) using a key
   C) Compresses it for storage
   D) Backs it up automatically
2. In symmetric encryption, how many keys are used to lock and unlock the data?
   A) Two different keys
   B) One shared key
   C) Three keys
   D) No key is needed
3. In asymmetric encryption, what is the purpose of the public key?
   A) It must be kept completely secret
   B) It can be shared with anyone and used to encrypt messages sent to the owner
   C) It decrypts all messages on the internet
   D) It is used only for hashing passwords
4. In asymmetric encryption, who can decrypt a message encrypted with a person's public key?
   A) Anyone on the internet
   B) Only the holder of the matching private key
   C) Any firewall
   D) Any antivirus program

5. What is the term for the large-scale system that manages public/private key pairs?
   A) IDS
   B) Public Key Infrastructure (PKI)
   C) DDoS protection
   D) Sandboxing framework
6. What makes hashing different from encryption?
   A) Hashing is reversible, encryption is not
   B) Hashing is a one-way transformation and cannot be reversed back to the original data
   C) Hashing requires a shared key
   D) Hashing only works on images
7. Why do secure websites store a hash of a password instead of the password itself?
   A) Hashes take up less storage space
   B) So that even if the database is stolen, the original password cannot be directly recovered
   C) Hashing makes login faster
   D) Hashing is required for two-factor authentication
8. What is the purpose of adding a "salt" before hashing a password?
   A) To make the hash shorter
   B) To add random extra data so stolen password databases are far harder to crack
   C) To make the password symmetric
   D) To convert the hash into a public key
9. The protocol HTTPS relies on which underlying concept to keep data secure during transmission?
   A) Sandboxing
   B) Encryption
   C) E-waste management
   D) Digital etiquette
10. What analogy best represents asymmetric encryption's public key?
   A) A shared house key copied for two people
   B) An open mailbox slot that anyone can drop a letter into
   C) A firewall rule list
   D) A backup hard drive

Topic 9.2.4: Detecting and Preventing Intrusions

1. What is the key difference between an IDS and an IPS?
   A) IDS blocks threats; IPS only detects them
   B) IDS only detects and alerts; IPS actively detects and blocks threats
   C) They are identical tools
   D) IPS only works on mobile devices

2. What is a "log" in the context of intrusion detection?
   A) A type of malware
   B) A record of every action taken on a system
   C) A public encryption key
   D) A type of firewall

3. What is an "anomaly" in system monitoring?
   A) A routine, expected login
   B) Activity that breaks the normal expected pattern, signaling possible intrusion
   C) A scheduled software update
   D) A properly hashed password

4. In a SQL Injection attack, what does the attacker typically manipulate?
   A) The device's hardware
   B) User input fields to alter a database query
   C) The Wi-Fi router firmware
   D) The antivirus signature database

5. What is the recommended fix for preventing SQL Injection attacks?
   A) Disabling all logins
   B) Using parameterized queries that treat user input strictly as data
   C) Removing the database entirely
   D) Turning off the firewall

6. Which of the following would most likely be flagged as a suspicious anomaly in system logs?
   A) A regular login from the user's home city during work hours
   B) A login from an unfamiliar country at 3 a.m. followed by a large file download
   C) A scheduled software update
   D) A user changing their profile picture

Topic 9.3.1: Designing for Equity and Accessibility

1. What does the "ramp beside the stairs" analogy represent in accessibility design?
   A) Making an app more expensive to build
   B) Adding accessible design that helps some users without disadvantaging others
   C) Removing all visual design elements
   D) Restricting access to fewer users

2. What does WCAG stand for?
   A) World Computer Access Guidelines
   B) Web Content Accessibility Guidelines
   C) Wireless Connectivity and Graphics
   D) Web Coding and Application Growth

3. In the POUR principles of WCAG, what does the "O" stand for?
   A) Optional
   B) Operable
   C) Outdated
   D) Original

4. What does "Perceivable" mean in the WCAG POUR framework?
   A) Content must load quickly
   B) Users must be able to perceive the content, e.g., through alt text or captions
   C) Content must be colorful
   D) Content must be short

5. What is "contrast ratio" used to measure in accessibility?
   A) Internet speed
   B) The difference between text color and background color for readability
   C) File compression level
   D) Password strength

6. According to WCAG, what is the minimum recommended contrast ratio for normal text?
   A) 1:1
   B) 4.5:1
   C) 10:1
   D) 100:1

7. Which WCAG principle ensures a website works reliably across different devices and assistive technologies?
   A) Perceivable
   B) Robust
   C) Understandable
   D) Operable only

Topic 9.3.2: Creating Inclusive Applications

1. What is "low-bandwidth optimization" designed to address?
   A) Screen reader compatibility
   B) Users with slow or limited internet access
   C) Password strength
   D) Multi-factor authentication

2. Why is multilingual support important for inclusive application design in Pakistan?
   A) It increases app file size
   B) It allows non-native English speakers and regional language users to access content
   C) It removes the need for accessibility features
   D) It replaces the need for privacy settings

3. Which design choice most directly benefits users with limited digital literacy?
   A) Complex, feature-dense menus
   B) Simple layouts with clear buttons and short instructions
   C) Removing all instructions
   D) Requiring advanced technical knowledge to navigate

4. What is a key benefit of adding descriptive alt text to images?
   A) It increases loading speed
   B) It allows screen reader users to understand image content
   C) It hides the image from search engines
   D) It reduces server costs

Topic 9.3.3: Strategies for Effective and Secure Remote Collaboration

1. What is the benefit of assigning clear roles in a remote team?
   A) It slows down the project
   B) It ensures no task is duplicated or forgotten
   C) It removes the need for communication
   D) It eliminates the need for deadlines

2. Which tool is commonly used to track progress in remote collaboration?
   A) Antivirus software
   B) A shared task board or checklist
   C) A firewall
   D) A hashing algorithm

3. Why should the same access-control habits from Section 9.1 be applied during remote collaboration?
   A) To make files load faster

B) To ensure only authorized people can view or edit shared work
C) To increase file size
D) To disable notifications

Topic 9.4.1: Characteristics of Successful Entrepreneurs

1. Which of the following best describes a "calculated risk," a trait of successful entrepreneurs?
   A) Avoiding all risk entirely
   B) Taking a risk after weighing potential outcomes rather than acting recklessly
   C) Risking money without any planning
   D) Copying a competitor exactly

2. Why is adaptability considered an important entrepreneurial trait?
   A) It allows entrepreneurs to ignore customer feedback
   B) It allows entrepreneurs to adjust quickly when a plan isn't working
   C) It removes the need for a business plan
   D) It guarantees business success

3. What mindset helps entrepreneurs make decisions despite incomplete information?
   A) Waiting indefinitely for certainty
   B) Making informed decisions with the information available
   C) Avoiding decision-making entirely
   D) Relying only on luck

Topic 9.4.2: Entrepreneurship in Pakistan and the Local Digital Economy

1. What has been a major driver of Pakistan's growing digital economy?
   A) Declining internet access
   B) Cheap mobile data and a young population
   C) Reduced use of smartphones
   D) Fewer freelancing platforms

2. What is the purpose of Roshan Digital Accounts?
   A) To block international payments
   B) To make it easier for overseas Pakistanis to move money digitally
   C) To replace freelancing platforms
   D) To restrict e-commerce growth

3. Which of the following is a widely used freelancing platform mentioned in the chapter?
   A) Daraz
   B) Fiverr
   C) VLC Player
   D) Notepad

Topic 9.4.3: Identifying Business Opportunities and Recognizing Market Needs

1. What is the first step in identifying a genuine business opportunity?
   A) Building the final product
   B) Observing a real, daily-life problem
   C) Setting a high price
   D) Hiring a large team

2. What does "willingness to pay" help an entrepreneur determine?
   A) The app's color scheme
   B) Whether customers value the solution enough to pay for it
   C) The number of competitors
   D) The internet speed required

3. What is the fastest way to validate whether a business problem is real before building a solution?
   A) Guessing based on personal opinion
   B) Surveys, social media comments, and direct customer conversations
   C) Skipping research and launching immediately
   D) Copying a random existing business

Topic 9.4.4: Digital Platforms for Entrepreneurship

1. Which platform type is best suited for selling physical products at scale?
   A) Freelancing platforms
   B) E-commerce platforms
   C) Social media only
   D) Antivirus software

2. Which platform type is best suited for selling skills and services globally?
   A) E-commerce platforms
   B) Freelancing platforms
   C) Firewalls
   D) IDS tools

3. Social media platforms are primarily useful for entrepreneurs in which way?
   A) Encrypting customer data
   B) Brand building and directly reaching customers
   C) Managing firewalls
   D) Filing PECA complaints

Topic 9.4.5: Freelancing and Remote Work Opportunities

1. Why is it risky for a freelancer to use informal, unverified payment methods?
   A) They are always faster
   B) They carry financial and legal risk compared to verified payment channels
   C) They are required by freelancing platforms
   D) They automatically increase income

2. What is the recommended first step when building a freelance portfolio?
   A) Add as many unrelated samples as possible
   B) Choose one specific, sellable skill and build a small, strong portfolio
   C) Set the highest possible price immediately
   D) Avoid client communication

3. What should a freelancer confirm with a client before starting work?
   A) Only the final price
   B) Scope, timeline, and revisions
   C) The client's personal address
   D) The client's browsing history

Topic 9.4.6: Planning a Small Digital Business

1. What does MVP stand for in digital business planning?
   A) Most Valuable Product
   B) Minimum Viable Product
   C) Maximum Value Proposition
   D) Multi-Vendor Platform

2. What is the purpose of building an MVP rather than a fully-featured product immediately?
   A) To impress investors with complexity
   B) To build the smallest working version that tests the core idea
   C) To avoid customer feedback
   D) To maximize initial cost

3. In the business planning pipeline, which stage comes immediately after Market Validation?
   A) User Acquisition
   B) MVP Development
   C) Feedback & Iteration
   D) Payment Integration
4. Why is a failed first version of a product considered valuable in entrepreneurship?
   A) It guarantees future failure
   B) It provides feedback and data for improvement, not defeat
   C) It has no impact on the business
   D) It should be ignored completely

Topic 9.4.7: Innovation, Emerging Technologies, and Career Pathways

1. How have emerging technologies like AI and cloud computing affected the cost of starting a business?
   A) They have significantly increased costs
   B) They have lowered the cost of starting a business
   C) They have made entrepreneurship illegal
   D) They have no effect on cost
2. Which of the following is a career pathway branching from digital entrepreneurship?
   A) E-waste collector only
   B) App developer
   C) Firewall hardware technician only
   D) None of the above

Topic 9.5.1: Ethical Use of Technology and Intellectual Property

1. What does "intellectual property" refer to?
   A) Physical computer hardware
   B) Creative work such as software, music, and writing that legally belongs to its creator
   C) A type of firewall
   D) A government tax policy
2. Using someone else's code from a forum without credit or permission is an example of:
   A) Ethical and legal technology use
   B) A violation of intellectual property and digital ethics
   C) Sustainable computing
   D) Accessibility design

3. Which of the following is an example of unethical digital behavior?
   A) Reporting a data breach
   B) Spreading false information or cyberbullying
   C) Using strong passwords
   D) Reading a privacy policy

Topic 9.5.2: Legal Aspects and Data Privacy Laws

1. What does GDPR stand for?
   A) Global Data Privacy Regulation
   B) General Data Protection Regulation
   C) Government Data Protection Rule
   D) General Digital Privacy Rights

2. GDPR is a data protection law associated with which region?
   A) Pakistan
   B) The European Union
   C) The United States only
   D) Africa

3. What does PECA stand for?
   A) Pakistan Electronic Communication Authority
   B) Prevention of Electronic Crimes Act
   C) Personal Encryption and Cyber Access
   D) Public Electronic Consumer Agreement

4. In what year was PECA enacted in Pakistan?
   A) 2010
   B) 2016
   C) 2020
   D) 1994

5. Which of the following does PECA criminalize?
   A) Using strong passwords
   B) Unauthorized access to systems, hacking, and online fraud
   C) Reading privacy policies
   D) Using two-factor authentication

6. Under GDPR, organizations must provide users with the right to:
   A) Sell user data freely
   B) Access or delete their own personal data
   C) Avoid all data collection laws
   D) Ignore user consent

Topic 9.5.3: Environmental and Social Impacts

1. What is "e-waste"?
   A) Biodegradable packaging
   B) Discarded electronic devices and their environmentally harmful materials
   C) Digital collaboration software
   D) A type of encryption

2. What environmental concern is associated with large data centers and cryptocurrency mining?
   A) Reduced internet speed
   B) High energy consumption
   C) Increased accessibility
   D) Improved data privacy

3. What does "sustainable computing" involve?
   A) Upgrading devices as frequently as possible
   B) Recycling responsibly and using energy-efficient devices
   C) Ignoring e-waste disposal
   D) Avoiding all software updates

4. Improper disposal of e-waste can result in:
   A) Improved soil quality
   B) Toxic materials leaking into soil and water
   C) Faster internet speeds
   D) Stronger encryption

Topic 9.5.4: The Role of Technology in Cultural and Societal Change

1. What term describes the gap between people with reliable internet access and those without?
   A) E-waste
   B) Digital divide
   C) Data breach
   D) Sandbox gap

2. Technology's influence on culture and society includes which of the following?
   A) No effect on communication
   B) Changes in how people communicate, learn, and access opportunity
   C) Complete elimination of privacy concerns
   D) Guaranteed equal access for all users

Section B: Short Answer Questions (SQs)

Topic 9.1.1: Software Maintenance and Security Patching Mechanics

1. Define the term "patch" in the context of software security.
2. Differentiate between a zero-day exploit and an N-day vulnerability.
3. Explain briefly why security researchers report flaws privately to companies before public disclosure.
4. List the sequence of steps from the discovery of a vulnerability to the closing of that security gap on a user's device.
5. Explain briefly how the Morris Worm incident changed the way the tech industry responds to security threats.
6. Predict what could happen to a device that consistently ignores software update notifications.

Topic 9.1.2: Privacy and Security Settings

1. Define "privacy policy" in your own words.
2. Differentiate between account visibility settings and app permission settings.
3. Explain briefly why default app settings often favor data collection over user privacy.
4. State two examples of privacy settings a user can customize on a social media platform.
5. Identify the risk of leaving location sharing turned on permanently for all apps.

Topic 9.1.3: Safe Online Collaboration Workflows

1. Define "access control" as it applies to shared digital files.
2. Differentiate between "View" and "Edit" permission levels in a shared document.
3. Explain briefly what a token is and why stealing one is dangerous for a user's account.
4. Diagram/map out the steps you would take to safely share a school project folder with three classmates.
5. Identify why setting a shared folder to "Anyone with the link can edit" is considered risky.
6. State two examples of digital etiquette that should be followed in a group project chat.

Topic 9.2.1: Understanding Cyber Threats

1. Define "phishing" and give one real-world style example.
2. Differentiate between ransomware and spyware.
3. Differentiate between a Man-in-the-Middle attack and a DDoS attack.
4. Explain briefly why public Wi-Fi networks increase the risk of Man-in-the-Middle attacks.
5. Explain briefly what a zero-day exploit is and why it is particularly dangerous.

6. Identify what the Stuxnet incident revealed about the real-world capabilities of malicious software.
7. Predict the likely outcome for a company whose servers experience a large-scale DDoS attack during business hours.

Topic 9.2.2: Security Tools and Techniques

1. Define "firewall" and state its main function.
2. Differentiate between an Intrusion Detection System (IDS) and antivirus software.
3. Explain briefly how Multi-Factor Authentication (MFA) reduces the risk of account takeover even if a password is stolen.
4. Define "sandboxing" and explain briefly why it is a useful security technique.
5. State two examples of biometric verification methods.
6. Diagram/map out how a firewall and IDS work together to protect a network, using the "gate guard and camera" analogy.

Topic 9.2.3: Data Protection and Cryptography

1. Define "encryption" and "decryption" in your own words.
2. Differentiate between symmetric and asymmetric encryption.
3. Explain briefly the role of a public key versus a private key in asymmetric encryption.
4. Define "hashing" and explain briefly why it cannot be reversed.
5. Explain briefly why adding a "salt" to a password before hashing improves security.
6. Diagram/map out the steps of how Alice would send Bob a secure message using asymmetric encryption.
7. Identify why storing plain-text passwords in a database is considered a serious security failure.

Topic 9.2.4: Detecting and Preventing Intrusions

1. Define "log" and explain briefly its role in detecting intrusions.
2. Differentiate between an IDS and an IPS.
3. Explain briefly what an "anomaly" is in the context of system monitoring.
4. Predict the output of a login query when an attacker inputs `' OR '1'='1` into an unprotected login form.
5. Explain briefly how a parameterized query prevents SQL injection attacks.
6. Identify two examples of suspicious activity that might appear in a system's access logs.

Topic 9.3.1: Designing for Equity and Accessibility

1. Define "accessibility" as it relates to digital design.

2. Explain briefly what each letter of the WCAG POUR framework stands for.
3. Define "contrast ratio" and explain briefly why it matters for users with low vision.
4. Explain briefly the "ramp beside the stairs" analogy and what it represents in accessible design.
5. Identify two features a website could add to become more "Perceivable" for visually impaired users.

Topic 9.3.2: Creating Inclusive Applications

1. Define "low-bandwidth optimization" and explain briefly why it matters in Pakistan.
2. State two features that make an application more inclusive for diverse users.
3. Explain briefly why multilingual support is important for digital equity.
4. Identify one design choice that could exclude users with limited digital literacy.

Topic 9.3.3: Strategies for Effective and Secure Remote Collaboration

1. State two strategies for effective remote team collaboration.
2. Explain briefly why assigning clear roles improves teamwork in digital projects.
3. Differentiate between a task board and a shared communication channel in a remote team's workflow.
4. Identify why access-control practices from Section 9.1 remain important during remote collaboration.

Topic 9.4.1: Characteristics of Successful Entrepreneurs

1. List three characteristics commonly shared by successful entrepreneurs.
2. Differentiate between a "calculated risk" and a reckless business decision.
3. Explain briefly why adaptability is important for entrepreneurial success.

Topic 9.4.2: Entrepreneurship in Pakistan and the Local Digital Economy

1. State two factors driving the growth of Pakistan's digital economy.
2. Define "Roshan Digital Accounts" and explain briefly their purpose.
3. Identify two digital platforms commonly used by entrepreneurs in Pakistan.

Topic 9.4.3: Identifying Business Opportunities and Recognizing Market Needs

1. Explain briefly how an entrepreneur can identify a genuine business opportunity.
2. Define "willingness to pay" and explain briefly its importance in market validation.
3. State two methods entrepreneurs use to validate a business idea before building it.

Topic 9.4.4: Digital Platforms for Entrepreneurship

1. Differentiate between e-commerce platforms and freelancing platforms.
2. Explain briefly how social media supports digital entrepreneurship.
3. State one example each of a social media platform, an e-commerce platform, and a freelancing platform.

Topic 9.4.5: Freelancing and Remote Work Opportunities

1. Define "freelancing" in the context of the digital economy.
2. Explain briefly why verified payment methods are important for freelancers.
3. State two things a freelancer should confirm with a client before beginning work.

Topic 9.4.6: Planning a Small Digital Business

1. Define "MVP (Minimum Viable Product)."
2. Diagram/map out the five-stage pipeline for planning a small digital business.
3. Explain briefly why "Feedback & Iteration" is an important final stage of the business pipeline.
4. Explain briefly why a failed first product version should be viewed as useful data.

Topic 9.4.7: Innovation, Emerging Technologies, and Career Pathways

1. State two emerging technologies transforming digital entrepreneurship.
2. Explain briefly how emerging technology has lowered the barrier to starting a business.
3. Identify two career pathways that stem from digital entrepreneurship.

Topic 9.5.1: Ethical Use of Technology and Intellectual Property

1. Define "intellectual property" and give one example.
2. Explain briefly why copying code without credit is considered both an ethical and legal issue.
3. State two examples of unethical digital behavior.

Topic 9.5.2: Legal Aspects and Data Privacy Laws

1. Define GDPR and state the region it applies to.
2. Define PECA and state the year it was enacted in Pakistan.
3. Differentiate between GDPR and PECA in terms of jurisdiction and scope.
4. Explain briefly what rights GDPR grants users regarding their personal data.
5. Identify two types of cybercrime covered under PECA.

Topic 9.5.3: Environmental and Social Impacts

1. Define "e-waste" and give two examples.

2. Explain briefly why data centers and cryptocurrency mining contribute to environmental strain.
3. Define "sustainable computing" and state two practices associated with it.

Topic 9.5.4: The Role of Technology in Cultural and Societal Change

1. Define "digital divide."
2. Explain briefly how technology has changed the way people communicate and learn.
3. Identify one way the digital divide could negatively affect a community.

Section C: Long / Extensive Questions (LQs)

Topic 9.1: Safe and Responsible Digital Practices

1. Analyze the complete lifecycle of a software vulnerability, from discovery to patching.
   a) Explain how a vulnerability is typically discovered and reported.
   b) Describe the process a company follows to build and release a patch.
   c) Discuss the risks a user faces during the "danger window" between public disclosure and personal patch installation.
   d) Justify why regular software updates should be treated as a security priority rather than an optional task.

2. Discuss in detail how privacy settings, access control, and permission levels work together to keep a shared academic project secure.
   a) Explain the role of privacy settings on personal accounts.
   b) Explain the role of access control and permission levels in shared collaboration tools.
   c) Design a complete workflow for safely creating, sharing, and eventually closing access to a group project folder.

3. Evaluate the real-world consequences of the 1988 Morris Worm incident and construct an argument for why its lessons remain relevant to software maintenance practices today.

Topic 9.2: Cybersecurity Awareness and Threat Management

1. Analyze the major categories of cyber threats discussed in this unit.
   a) Define and explain phishing, ransomware, spyware, Man-in-the-Middle attacks, DDoS attacks, and zero-day exploits.
   b) Discuss how each threat type affects individuals differently from how it affects organizations.
   c) Evaluate which of these threats would be most difficult for an average user to detect, and justify your answer.

2. Construct a comparative analysis of the core security tools covered in this unit: firewalls, antivirus software, IDS, MFA, and sandboxing.
   a) Explain the specific function of each tool.
   b) Evaluate how these tools work together in a layered defense strategy.
   c) Design a basic security setup for a small business's office network using at least four of these tools, and justify each choice.

3. Elaborate on the concept of cryptography as a data protection method.
   a) Differentiate between symmetric and asymmetric encryption, including a full trace of how Alice would send Bob a secure message using asymmetric encryption.
   b) Explain the role of hashing and salting in protecting stored passwords, and justify why hashing is preferred over storing plain-text passwords.
   c) Discuss the real-world importance of Public Key Infrastructure (PKI) in enabling secure online communication, such as HTTPS and online banking.

4. Design and trace a complete SQL Injection attack and its defense.
   a) Construct the vulnerable query used in a basic login form.
   b) Trace step-by-step how an attacker's malicious input bypasses authentication.
   c) Construct the corrected, parameterized version of the query.
   d) Justify why parameterized queries prevent this class of attack.

5. Discuss in detail the difference between an Intrusion Detection System (IDS) and an Intrusion Prevention System (IPS), and evaluate the role of log analysis and anomaly detection in identifying a cyberattack in progress.

Topic 9.3: Digital Collaboration and Equity in Computing

1. Evaluate the importance of accessibility in modern application design.
   a) Explain each principle of the WCAG POUR framework in detail with an example for each.
   b) Discuss the real-world impact of poor accessibility design on users with different abilities.
   c) Design a short accessibility audit checklist that a student developer could use to test their own website.

2. Analyze the concept of digital equity in the context of Pakistan's diverse user base.
   a) Discuss the challenges posed by limited internet bandwidth, language barriers, and low digital literacy.
   b) Propose at least three specific design strategies to make a digital application more inclusive for these users.
   c) Evaluate the trade-offs a developer might face between adding rich features and maintaining accessibility/performance for low-bandwidth users.

3. Discuss in detail the strategies required for effective and secure remote team collaboration, and construct a complete workflow plan for a five-person student team completing a semester-long digital project, including role assignment, task management, communication rules, and security practices.

Topic 9.4: Digital Entrepreneurship and Innovation

1. Elaborate on the concept of digital entrepreneurship and its growth within Pakistan's economy.
   a) Discuss the key characteristics of successful entrepreneurs.
   b) Analyze the factors driving Pakistan's growing digital economy, including the role of initiatives like Roshan Digital Accounts.
   c) Evaluate at least three digital platforms (social media, e-commerce, freelancing) and discuss which is best suited for different types of entrepreneurs.
2. Design a complete business plan for a small digital venture of your choice.
   a) Identify a real-world problem and validate the business opportunity.
   b) Construct the full business pipeline: Market Validation, MVP Development, Payment Integration, User Acquisition, and Feedback & Iteration.
   c) Discuss a basic budgeting and resource plan for the venture.
   d) Justify your team structure and communication plan.
3. Analyze the freelancing and remote work landscape in Pakistan.
   a) Discuss the steps required to build a successful freelance pipeline, from portfolio creation to client payment.
   b) Evaluate the risks associated with informal or unverified payment methods compared to verified platform payouts.
   c) Discuss how emerging technologies such as AI and cloud computing are changing career opportunities for young freelancers and entrepreneurs.
4. Case Study: A group of students wants to launch an online organic-produce delivery app for a major Pakistani city.
   a) Identify at least four unique local challenges (payment, logistics, trust, connectivity) the team must solve.
   b) Propose specific solutions for each challenge.
   c) Evaluate how the business could apply the Market Validation → MVP → Payment Integration → User Acquisition → Feedback pipeline to this specific idea.

Topic 9.5: Ethical, Legal, and Environmental Impacts of Computing

1. Discuss in detail the ethical responsibilities of technology users and creators.
   a) Explain the concept of intellectual property and why it must be respected.
   b) Discuss the consequences of unethical digital behavior such as hacking, spreading misinformation, and cyberbullying.
   c) Evaluate a scenario where a student copies another developer's code without credit, and discuss both the ethical and legal implications.

2. Analyze the legal frameworks that govern data privacy and cybercrime.
   a) Explain the core principles and requirements of the GDPR.
   b) Explain the scope and purpose of Pakistan's PECA (2016), including the types of cybercrime it covers.
   c) Compare and contrast GDPR and PECA in terms of jurisdiction, enforcement, and user rights.
   d) Evaluate the legal and ethical risks faced by a company that collects and sells user location data without proper consent, and suffers a subsequent data breach.

3. Discuss in detail the environmental and social impacts of computing.
   a) Explain the concept of e-waste and its effect on the environment.
   b) Discuss the environmental cost of data centers and cryptocurrency mining.
   c) Propose at least three practical sustainable computing practices that individuals and organizations can adopt.
   d) Evaluate the role of technology in creating and widening the digital divide, and discuss what steps could help close that gap.

4. Prove, using evidence from the unit, that Stuxnet represented a turning point in how cybersecurity threats are understood, and evaluate the broader implications of software being able to cause physical, real-world damage.